Supplementary Material

When Accuracy Is Not Understanding: Auditing Annotation Shortcuts in a ClinVar–gnomAD Benchmark for AI-Assisted Variant Interpretation

Contents

Table S1 fold-level AUROC; Table S2 probe regularisation; Table S3 ClinVar review-status sensitivity; Table S4 all annotation strata; Table S5 incremental value of the reference-only score; Table S6 consequence-term counts; Section S7 reproducibility details and analyses not included.

Feature source	Group 1	Group 2	Group 3
Annotation: consequence terms + substitution	0.969	0.970	0.965
Annotation: harmonised flags (2)	0.873	0.828	0.858
Annotation: plain-missense flag only	0.854	0.820	0.838
DNABERT-2, reference-only	0.682	0.691	0.685
DNABERT-2, variant (Δ)	0.518	0.511	0.521
HyenaDNA, reference-only	0.620	0.619	0.647
HyenaDNA, variant (Δ)	0.546	0.526	0.550
4-mer, reference-only	0.566	0.579	0.582
4-mer, variant (Δ)	0.507	0.534	0.512
Substitution type only (no annotation)	0.586	0.613	0.600

Supplementary Table S1: AUROC within each held-out chromosome group (pooled out-of-fold predictions split by group). Groups follow the definitions of Feng et al. (2025): group 1 = chr3, 6, 9, 12, 16, 18, 19, 21; group 2 = chr2, 5, 11, 14, 17, 20, 22, X; group 3 = chr1, 4, 7, 8, 10, 13, 15.

S7. Reproducibility details and analyses not included

Software and settings. Python with numpy, pandas, SciPy, scikit-learn (logistic regression, L-BFGS; StandardScaler fitted on training folds) and matplotlib; random seed 42 for sampling, resampling and probes. Versions: Python 3.13, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, scikit-learn 1.9.1, Matplotlib 3.11.2; `alphagenome-pytorch` 0.3.1 (embeddings computed earlier).

Files. The analysis folder contains `scripts/01_leakage_main.py` (annotation-only and embedding probes, stratified analysis, incremental value, label-shuffle control), `02_crosstab_and_alphagenome.py` (vocabulary cross-tabulation; AlphaGenome subset), `03_shared_vocab_and_figure.py` (shared-vocabulary probe; Figure 2), `04_make_manuscript_assets.py` and `05_extra_assets.py` (robustness analyses and every table and number in the manuscript), and `build_manuscript.sh`. Result tables are in `results/`. Per-variant out-of-fold scores are in `results/oof_scores_main.csv`.

Data sources. Variants and annotation: public benchmark of Feng et al. (2025) and its annotation table. Embeddings: DNABERT-2, HyenaDNA and a 4-mer baseline (501-bp window,

Feature source	C	Reference-only AUROC	Variant (Δ) AUROC
DNABERT-2	0.001	0.712	0.502
DNABERT-2	0.01	0.708	0.504
DNABERT-2	0.1	0.691	0.506
DNABERT-2	1	0.681	0.507
DNABERT-2	10	0.680	0.507
HyenaDNA	0.001	0.556	0.511
HyenaDNA	0.01	0.593	0.525
HyenaDNA	0.1	0.612	0.527
HyenaDNA	1	0.617	0.524
HyenaDNA	10	0.616	0.523
4-mer	0.001	0.573	0.496
4-mer	0.01	0.574	0.504
4-mer	0.1	0.572	0.504
4-mer	1	0.568	0.504
4-mer	10	0.564	0.504
Annotation (terms + substitution)	0.001	–	0.964
Annotation (terms + substitution)	0.01	–	0.968
Annotation (terms + substitution)	0.1	–	0.968
Annotation (terms + substitution)	1	–	0.968
Annotation (terms + substitution)	10	–	0.967

Supplementary Table S2: AUROC across L2 regularisation strengths C (inverse regularisation; larger is weaker). The main text uses $C = 1$, fixed in advance and identical for all feature sources.

3,000 variants per class, seed 42), and AlphaGenome embeddings for the 1,018-variant upstream test split. All embeddings were computed before this analysis.

Counts. The annotation table contains 22,254 pathogenic and 17,396 unique common variants after removing duplicate coordinates; this differs by a few rows from the 39,637 variants in the source sequence table. The sampled 6,000 variants were all matched to annotation.

Analyses not included. An earlier version of the project compared seven models with pairwise significance tests and an embedding-geometry analysis (principal components, silhouette, k-nearest-neighbour purity, centred kernel alignment, UMAP). These were removed because the study's question concerns benchmark construction, not model ranking, and because variant-specific results were uniformly close to chance. The relevant models were not rerun.

Not done. Pooling at the variant position, intermediate-layer analyses, longer-window replication, gene-held-out evaluation and any ancestry-stratified analysis were not performed.

Pathogenic subset (pathogenic / common)	Feature source	AUROC (95% CI)
All variants (3,000 / 3,000)		
Annotation: consequence terms + substitution		0.968 (0.963–0.972)
DNABERT-2, reference-only		0.681 (0.668–0.694)
DNABERT-2, variant (Δ)		0.507 (0.492–0.522)
HyenaDNA, reference-only		0.617 (0.604–0.632)
Pathogenic with assertion criteria (1,763 / 3,000)		
Annotation: consequence terms + substitution		0.968 (0.963–0.972)
DNABERT-2, reference-only		0.690 (0.675–0.705)
DNABERT-2, variant (Δ)		0.512 (0.495–0.530)
HyenaDNA, reference-only		0.614 (0.598–0.629)
Pathogenic multi-submitter/expert (554 / 3,000)		
Annotation: consequence terms + substitution		0.965 (0.959–0.969)
DNABERT-2, reference-only		0.689 (0.667–0.711)
DNABERT-2, variant (Δ)		0.491 (0.464–0.518)
HyenaDNA, reference-only		0.620 (0.594–0.644)

Supplementary Table S3: AUROC after restricting the pathogenic class by ClinVar review status (all common variants retained). The first subset is the full sample. Annotation-only AUROC is essentially unchanged, so the findings are not driven by the weakest assertions.

Stratum (pathogenic / common)	Feature source	AUROC (95% CI)	Probe trained
Plain missense only (2,247 / 236)			
	HyenaDNA reference-only	0.635 (0.596–0.674)	all variants
	HyenaDNA variant (Δ)	0.527 (0.487–0.563)	all variants
	DNABERT-2 reference-only	0.749 (0.711–0.782)	all variants
	DNABERT-2 variant (Δ)	0.493 (0.455–0.532)	all variants
	4-mer reference-only	0.552 (0.512–0.591)	all variants
	4-mer variant (Δ)	0.460 (0.422–0.501)	all variants
	Annotation	0.582 (0.545–0.620)	all variants
	HyenaDNA reference-only	0.649 (0.609–0.688)	within stratum
	DNABERT-2 reference-only	0.721 (0.683–0.757)	within stratum
Non-coding-transcript flagged (424 / 1,897)			
	HyenaDNA reference-only	0.585 (0.556–0.614)	all variants
	HyenaDNA variant (Δ)	0.509 (0.478–0.540)	all variants
	DNABERT-2 reference-only	0.633 (0.607–0.662)	all variants
	DNABERT-2 variant (Δ)	0.502 (0.472–0.532)	all variants
	4-mer reference-only	0.553 (0.523–0.583)	all variants
	4-mer variant (Δ)	0.512 (0.483–0.542)	all variants
	Annotation	1.000 (1.000–1.000)	all variants
	HyenaDNA reference-only	0.561 (0.534–0.591)	within stratum
	DNABERT-2 reference-only	0.560 (0.528–0.587)	within stratum
Other annotated (329 / 867)			
	HyenaDNA reference-only	0.604 (0.569–0.640)	all variants
	HyenaDNA variant (Δ)	0.559 (0.522–0.594)	all variants
	DNABERT-2 reference-only	0.611 (0.577–0.644)	all variants
	DNABERT-2 variant (Δ)	0.565 (0.527–0.601)	all variants
	4-mer reference-only	0.549 (0.514–0.586)	all variants
	4-mer variant (Δ)	0.563 (0.523–0.604)	all variants
	Annotation	0.976 (0.969–0.984)	all variants
	HyenaDNA reference-only	0.574 (0.539–0.610)	within stratum
	DNABERT-2 reference-only	0.566 (0.531–0.605)	within stratum

Supplementary Table S4: All annotation strata. "Annotation" within the non-coding-transcript stratum reaches AUROC 1.000 because the two classes spell the non-coding term differently; this is a vocabulary artefact, not a biological signal. Strata are defined from the consequence string: plain missense; any non-coding-transcript term; other.

Model	Annotation alone	+ reference-only score	Gain (95% CI)
HyenaDNA	0.968	0.972	0.005 (0.002–0.007)
DNABERT-2	0.968	0.977	0.010 (0.007–0.013)
4-mer	0.968	0.968	0.001 (-0.001–0.002)

Supplementary Table S5: Does the reference-only embedding score add information beyond annotation (consequence terms plus substitution type)? Gain is the paired-bootstrap difference in AUROC between the combined model and annotation alone. The gains are statistically distinguishable from zero for DNABERT-2 and HyenaDNA but small.

Consequence term	Pathogenic (n)	Common (n)	Total
missense_variant	22,254	16,557	38,811
non_coding_transcript_exon_variant	0	8,339	8,339
downstream_gene_variant	0	7,779	7,779
intron_variant	1,702	5,325	7,027
upstream_gene_variant	0	6,959	6,959
regulatory_region_variant	0	5,247	5,247
intron_variant&non_coding_transcript_variant	0	4,736	4,736
non-coding_transcript_variant	2,996	0	2,996
5_prime_UTR_variant	930	1,571	2,501
3_prime_UTR_variant&NMD_transcript_variant	0	2,289	2,289
missense_variant&NMD_transcript_variant	0	2,220	2,220
intron_variant&NMD_transcript_variant	0	1,916	1,916
3_prime_UTR_variant	271	1,012	1,283
synonymous_variant	64	723	787
TF_binding_site_variant	0	732	732
initiatior_codon_variant	663	0	663
missense_variant&splice_region_variant	0	383	383
splice_region_variant&non_coding_transcript_exon_variant	0	220	220
synonymous_variant&NMD_transcript_variant	0	181	181
nonsense	104	0	104

Supplementary Table S6: The twenty most frequent consequence terms (counts of variants carrying the term) in the full annotation table. A variant can carry several terms.